

LANDFORMS OF GLACIATION / HIMALAYAN GLACIERS-GLOF

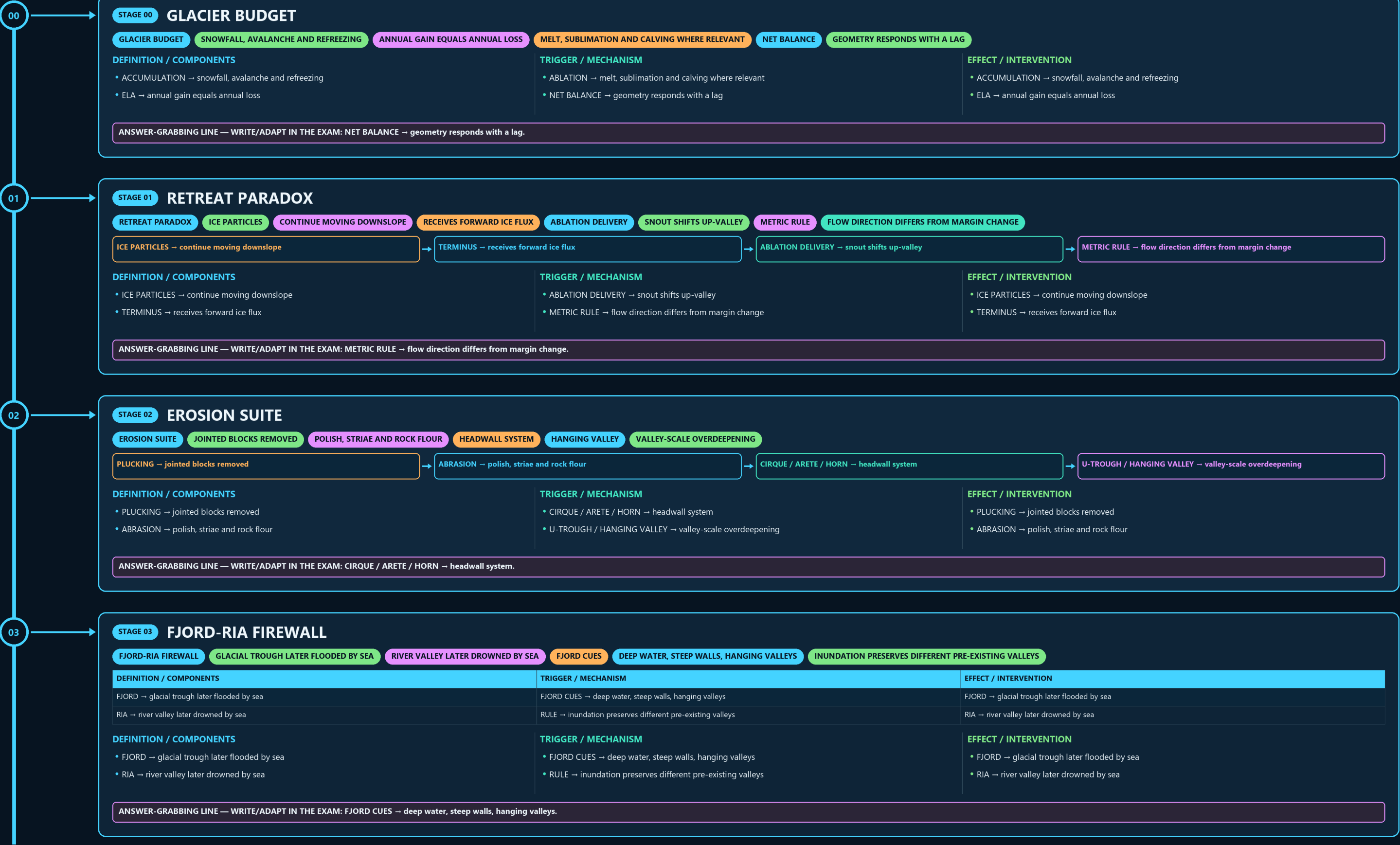
GLACIER BUDGET → RETREAT PARADOX → EROSION SUITE → FJORD-RIA FIREWALL → DEPOSITIONAL SORTING → ICE-MASS FAMILY

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g3 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



03

STAGE 03

FJORD-RIA FIREWALL

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GLACIAL TROUGH LATER FLOODED BY SEA

RIVER VALLEY LATER DROWNED BY SEA

FJORD CUES

DEEP WATER, STEEP WALLS, HANGING VALLEYS

INUNDATION PRESERVES DIFFERENT PRE-EXISTING VALLEYS

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
FJORD → glacial trough later flooded by sea	FJORD CUES → deep water, steep walls, hanging valleys	FJORD → glacial trough later flooded by sea
RIA → river valley later drowned by sea	RULE → inundation preserves different pre-existing valleys	RIA → river valley later drowned by sea

DEFINITION / COMPONENTS

- FJORD → glacial trough later flooded by sea
- RIA → river valley later drowned by sea

TRIGGER / MECHANISM

- FJORD CUES → deep water, steep walls, hanging valleys
- RULE → inundation preserves different pre-existing valleys

EFFECT / INTERVENTION

- FJORD → glacial trough later flooded by sea
- RIA → river valley later drowned by sea

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FJORD CUES → deep water, steep walls, hanging valleys.

04

STAGE 04

DEPOSITIONAL SORTING

DEPOSITIONAL SORTING

DIRECT ICE, GENERALLY UNSORTED

MELTWATER RIDGE IN OR UNDER ICE

BRAIDED MELTWATER, RELATIVELY SORTED

STREAMLINED DRIFT/TILL LANDFORM

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
TILL / MORaine → direct ice, generally unsorted	OUTWASH → braided meltwater, relatively sorted	TILL / MORaine → direct ice, generally unsorted
ESKER → meltwater ridge in or under ice	DRUMLIN → streamlined drift/till landform	ESKER → meltwater ridge in or under ice

DEFINITION / COMPONENTS

- TILL / MORaine → direct ice, generally unsorted
- ESKER → meltwater ridge in or under ice

TRIGGER / MECHANISM

- OUTWASH → braided meltwater, relatively sorted
- DRUMLIN → streamlined drift/till landform

EFFECT / INTERVENTION

- TILL / MORaine → direct ice, generally unsorted
- ESKER → meltwater ridge in or under ice

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: ESKER → meltwater ridge in or under ice.

05

STAGE 05

ICE-MASS FAMILY

ICE-MASS FAMILY

GLACIER, ICE CAP, ICE SHEET

SHELF, ICEBERG AND SEA ICE

SEA LEVEL

GROUNDING LOSS ADDS OCEAN MASS

SHELF LOSS CAN ACCELERATE INLAND DISCHARGE

FLOATING → shelf, iceberg and sea ice

GROUNDING → glacier, ice cap, ice sheet

SEA LEVEL → grounding loss adds ocean mass

BUTTRESSING → shelf loss can accelerate inland discharge

DEFINITION / COMPONENTS	TRIGGER / MECHANISM	EFFECT / INTERVENTION
- GROUNDING → glacier, ice cap, ice sheet - FLOATING → shelf, iceberg and sea ice	- SEA LEVEL → grounding loss adds ocean mass - BUTTRESSING → shelf loss can accelerate inland discharge	- GROUNDING → glacier, ice cap, ice sheet - FLOATING → shelf, iceberg and sea ice

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BUTTRESSING → shelf loss can accelerate inland discharge.

06

STAGE 06

HIMALAYAN MAP HOOKS

HIMALAYAN MAP HOOKS

BARA SHIGRI

CHENAB SYSTEM

NUBRA-SHYOK-INDUS SYSTEM

ZEMU → Teesta

GANGOTRI → Bhagirathi

BARA SHIGRI → Chenab system

SIACHEN → Nubra-Shyok-Indus system

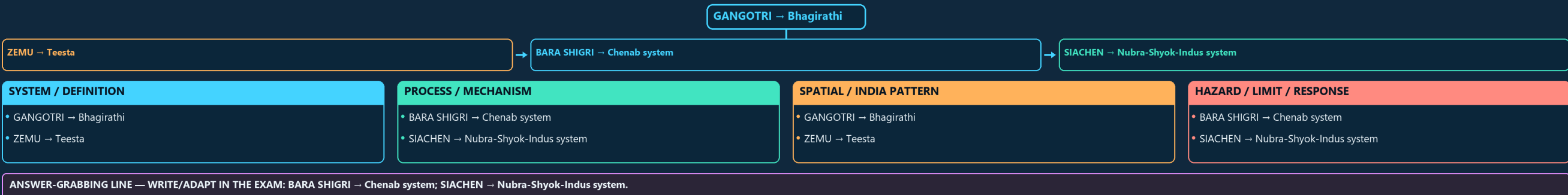
SYSTEM / DEFINITION	PROCESS / MECHANISM	SPATIAL / INDIA PATTERN	HAZARD / LIMIT / RESPONSE
- GANGOTRI → Bhagirathi - ZEMU → Teesta	- BARA SHIGRI → Chenab system - SIACHEN → Nubra-Shyok-Indus system	- GANGOTRI → Bhagirathi - ZEMU → Teesta	- BARA SHIGRI → Chenab system - SIACHEN → Nubra-Shyok-Indus system

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BARA SHIGRI → Chenab system; SIACHEN → Nubra-Shyok-Indus system.

06

STAGE 06 HIMALAYAN MAP HOOKS

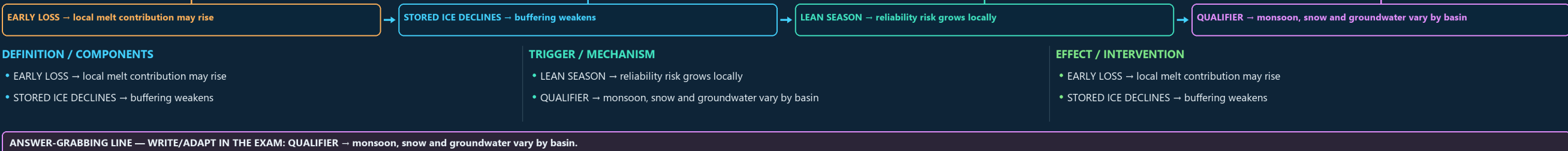
HIMALAYAN MAP HOOKS BARA SHIGRI CHENAB SYSTEM NUBRA-SHYOK-INDUS SYSTEM



07

STAGE 07 WATER-TIMING SEQUENCE

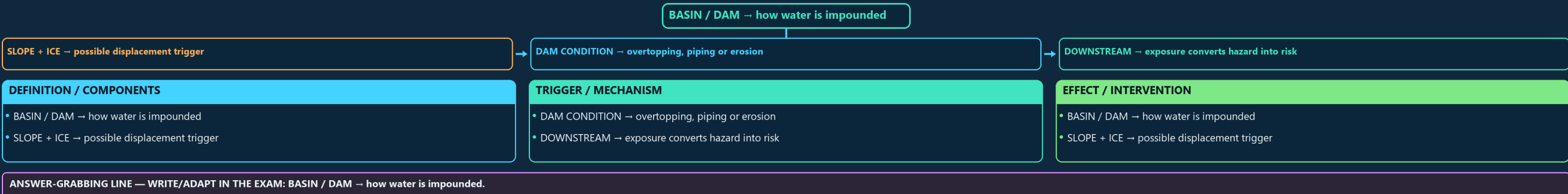
WATER-TIMING SEQUENCE EARLY LOSS LOCAL MELT CONTRIBUTION MAY RISE STORED ICE DECLINES BUFFERING WEAKENS LEAN SEASON RELIABILITY RISK GROWS LOCALLY MONSOON, SNOW AND GROUNDWATER VARY BY BASIN



08

STAGE 08 GLACIAL-LAKE TEST

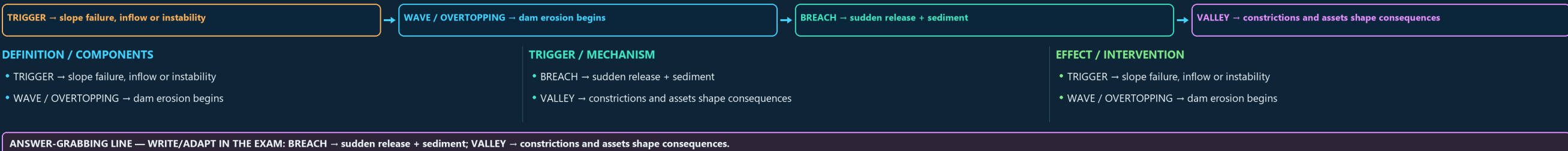
GLACIAL-LAKE TEST HOW WATER IS IMPOUNDED SLOPE + ICE POSSIBLE DISPLACEMENT TRIGGER DAM CONDITION OVERTOPPING, PIPING OR EROSION EXPOSURE CONVERTS HAZARD INTO RISK



09

STAGE 09 GLOF CASCADE

GLOF CASCADE SLOPE FAILURE, INFLOW OR INSTABILITY DAM EROSION BEGINS SUDDEN RELEASE + SEDIMENT CONSTRICTIONS AND ASSETS SHAPE CONSEQUENCES



09

STAGE 09 GLOF CASCADE

GLOF CASCADE SLOPE FAILURE, INFLOW OR INSTABILITY DAM EROSION BEGINS SUDDEN RELEASE + SEDIMENT CONSTRICTIONS AND ASSETS SHAPE CONSEQUENCES



DEFINITION / COMPONENTS

- TRIGGER → slope failure, inflow or instability
- WAVE / OVERTOPPING → dam erosion begins

TRIGGER / MECHANISM

- BREACH → sudden release + sediment
- VALLEY → constrictions and assets shape consequences

EFFECT / INTERVENTION

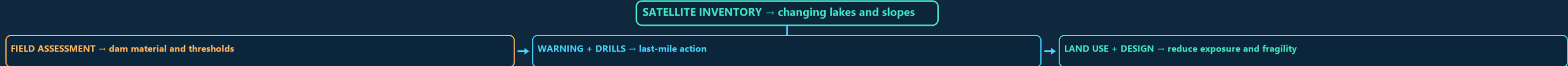
- TRIGGER → slope failure, inflow or instability
- WAVE / OVERTOPPING → dam erosion begins

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: BREACH → sudden release + sediment; VALLEY → constrictions and assets shape consequences.

10

STAGE 10 RISK-REDUCTION STACK

RISK-REDUCTION STACK SATELLITE INVENTORY CHANGING LAKES AND SLOPES FIELD ASSESSMENT DAM MATERIAL AND THRESHOLDS WARNING + DRILLS LAST-MILE ACTION LAND USE + DESIGN



DEFINITION / COMPONENTS

- SATELLITE INVENTORY → changing lakes and slopes
- FIELD ASSESSMENT → dam material and thresholds

TRIGGER / MECHANISM

- WARNING + DRILLS → last-mile action
- LAND USE + DESIGN → reduce exposure and fragility

EFFECT / INTERVENTION

- SATELLITE INVENTORY → changing lakes and slopes
- FIELD ASSESSMENT → dam material and thresholds

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: LAND USE + DESIGN → reduce exposure and fragility.

11

STAGE 11 CRYOSPHERE ANSWER SPINE

CRYOSPHERE ANSWER SPINE GLACIER, LAKE, GLOF AND RISK GLACIER-BASIN RELATION BUDGET, TRIGGER AND BREACH HETEROGENEITY, ATTRIBUTION AND DATA DATE



DEFINITION / COMPONENTS

- DEFINE → glacier, lake, GLOF and risk
- MAP → glacier-basin relation

TRIGGER / MECHANISM

- EXPLAIN → budget, trigger and breach
- QUALIFY → heterogeneity, attribution and data date

EFFECT / INTERVENTION

- DEFINE → glacier, lake, GLOF and risk
- MAP → glacier-basin relation

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: EXPLAIN → budget, trigger and breach; QUALIFY → heterogeneity, attribution and data date.

E

EXTRA SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT ABOUT 15,000 GLACIERS IN THE HIMALAYA, BETWEEN THE TWO LOWEST LEVEL OF PERPETUAL SNOW; IT VARIES WITH LATITUDE GREATER SNOWFALL/ACCUMULATION CAN LOWER THE CLIMATIC SNOWLINE, WHILE ARIDITY MAIN GLACIERS LIE IN THE GREATER AND TRANS-HIMALAYA (KARAKORAM) THE LESSER HIMALAYA HAS ONLY SMALL GLACIERS (TRACES IN) SIACHEN LIES IN THE EASTERN KARAKORAM UNDER INDIAN CONTROL. SOUTH LHONAK IS A MORaine-DAMMED GLACIAL LAKE THAT EXPANDED

OPTIONAL PROCESS DEPTH

- Khullar: about 15,000 glaciers in the Himalaya, between the two syntactical bends (east & west).
- Snowline = lowest level of perpetual snow; it varies with latitude, precipitation and local topography.
- Greater snowfall/accumulation can lower the climatic snowline, while aridity raises it; aspect,
- Main glaciers lie in the Greater and Trans-Himalaya (Karakoram, Ladakh, Zaskar)

DATA / INSTITUTION LIMIT

- the Lesser Himalaya has only small glaciers (traces in Pir-Panjal, Dhauladhar).
- Siachen lies in the eastern Karakoram under Indian control. Biafo, Baltoro and Hispar are in the
- South Lhonak is a moraine-dammed glacial lake that expanded as its glacier retreated; use a named
- The October 2023 flood involved lake outburst/overtopping after intense precipitation and a

CONTEMPORARY PARALLEL

- Mitigation: glacial-lake monitoring, early-warning systems, and restricting development in the flood path (NDMA GLOF guidelines).
- Khullar's ~15,000 figure is a source-era Himalayan estimate; modern inventory counts are method-dependent.
- Snowline controls: latitude, precipitation, topography.
- More accumulation generally lowers snowline; exact north-south contrast varies by Himalayan sector.

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.